

1.1 Propositional Logic

命題是陳述句，且一定有真OR假，不能同時都是真和假。

- Quick examples

" $1 + 1 = 2$ " $\rightarrow$ proposition

"What time is it?" $\rightarrow$ not a proposition

"Read this carefully." $\rightarrow$ not a proposition

$x + 1 = 2 \rightarrow$ not a proposition

- Atomic propositions

不能再拆成更簡單命題。

cannot be broken into simpler propositions.

- Compound propositions

由較簡單命題加上邏輯運算子組成。

built from simpler propositions using logical operators

- Logical operators

- (1) Negation $\neg p$

$\neg p$ means "not p."

It flips the truth value.

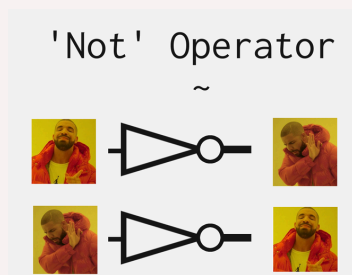
- Quick examples

p: The earth is round.

$\neg p$: The earth is not round.

*Other notations you might see are

- p , $\neg p$, $\bar{p}$, Np , and $!p$.

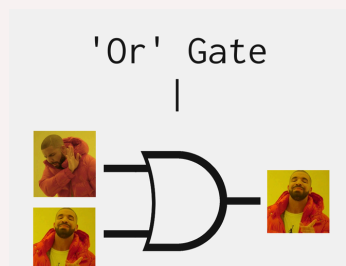


p	$\neg p$
T	F
F	T

- (2) Conjunction $p \wedge q$

$p \wedge q$ means "p and q."

It is true only when both are true.



p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

- (3) Disjunction $p \vee q$

$p \vee q$ means "p or q."

It is false only when both are false.

this is usually inclusive or.

Inclusive or = one is true, or both are true.

Exclusive or (XOR) = exactly one is true.



p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F